

Markscheme

Mathematics: analysis and approaches
Practice question bank

80. (a)

The first term of an arithmetic sequence is 6, and the sum of its first 10 terms is 195. A geometric sequence has the same first term as this arithmetic sequence, and the same 4th term.

Find the common ratio of the geometric sequence, correct to 3 significant figures.

$$S_{10} = \frac{10}{2}(2(6) + 9d) = 195 \quad (M1)$$

$$d = 3 \quad \mathbf{A1}$$

$$\text{4th term of the arithmetic sequence: } u_4 = 6 + 3(3) = 15 \quad (M1)$$

$$r^3 = \frac{15}{6} = 2.5 \Rightarrow r = 2.5^{\frac{1}{3}} \approx 1.357 \quad \mathbf{A1}$$

[6 marks]